GaN Building Block for a 1200 V, 300 kW Stacked Polyphase Bridge Inverter

From design inputs to the Rev A board: a decision-driven design

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Design flow

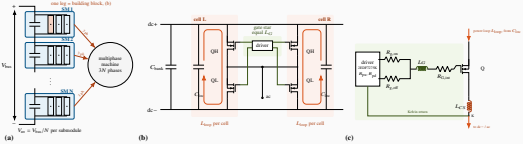


1	Design inputs	1200 V, 300 kW, $M \leq 1.15$, PMSM $\Rightarrow$ phase current
2	Device	survey 100–650 V Si/GaN $\Rightarrow$ 100 V GaN, EPC2361
3	Submodule voltage	75 V: fewest submodules within the 100 V rating $\Rightarrow$ $N = 16$
4	One device, one driver	Q3D cell + LTspice: how fast, how much overshoot, which losses
5	Devices in parallel	identical cells: loss, T_j, V_{DS} vs n $\Rightarrow$ at least 4 per switch
6	Gate drive	one driver for many devices breaks dynamic sharing $\Rightarrow$ one driver per two cells = building block
7	Two blocks	block-to-block sharing and circulating current via the AC connection
8–9	Verification, Rev A	DC link, parasitics, losses, thermal, gate bump, board, measurements

1 Design inputs

Design inputs

Inputs → Device → Submodule voltage → One device, one driver → n in parallel → Gate drive → block → Two blocks → Verify block → Rev A board



DC bus	1200 V nominal
Output power	300 kW
Modulation index	$M \leq 1.15$ (SVM)
Machine	PMSM, $\cos\varphi$ assumed 0.9
Switching frequency	50 kHz
Coolant	60 °C, $T_j \leq 125$ °C
Device stress	$V_{DS} \leq 90$ V at nominal bus (100

SPB: N three-phase submodules in series on the DC side, each feeding one winding set; $V_m = 1200\,\mathrm{V}/N$.

Phase current follows from the inputs

Each of the $3N$ legs delivers $P/3N$ at the rms phase voltage $V_{\mathrm{ph}} = MV_m/(2\sqrt{2})$:

$$I_{\mathrm{rms}} = \frac{P}{3N\,V_{\mathrm{ph}}\cos\varphi} = \frac{P}{3\,\dfrac{M\,V_{\mathrm{bus}}}{2\sqrt{2}}\cos\varphi}$$

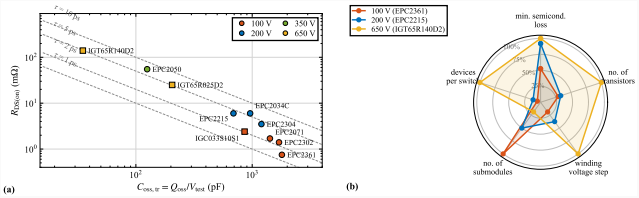
- $NV_m = V_{\mathrm{bus}}$: the phase current does not depend on N
- $V_{\mathrm{ph}} = 1.15 \cdot 75\,\mathrm{V}/2\sqrt{2} = 30.5\,\mathrm{V}$; 6.25 kW per leg
- **228 $\mathbf{A}_{\mathrm{rms}}$ per leg**, 322 A peak ($\cos\varphi = 0.9$)

$\cos\varphi$	I_{rms} per leg
1.00	205 A
0.95	216 A
0.90	**228 A**
0.85	241 A

Losses below are for $\cos\varphi = 0.9$; the PMSM operating point (field weakening, $\cos\varphi$) must be confirmed with the machine design.

2 Device

Voltage class: why 100 V GaN



- At fixed bus voltage and phase current the minimum semiconductor loss scales with the device time constant $R_{\mathrm{DS(on)}}\,C_{\mathrm{oss,tr}}$
- 100 V GaN gives the lowest product among 100–650 V Si/SiC/GaN candidates
- Selected: **EPC2361**, 100 V, 0.75 mΩ typ, 3×5 mm, top-side cooling

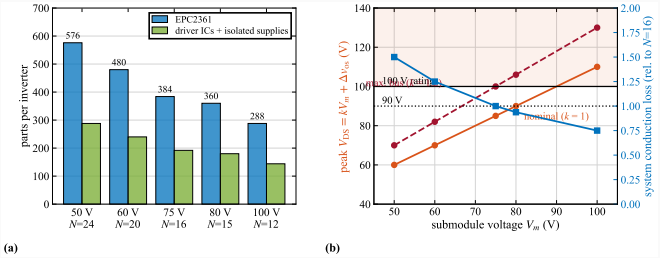
EPC2361 key data

V_{DS} / transient	100 / 120 V
$R_{\mathrm{DS(on)}}$	0.75 mΩ typ (1.0 max)
V_{GS} range	$-4 \ldots +6$ V
$V_{\mathrm{GS(th)}}$	0.8 / 1.1 / 2.5 V
C_{iss} / C_{oss} / C_{rss}	3599 / 1019 / 13 pF
Q_G / Q_{GD} / Q_{OSS}	28 / 3.8 / 90 nC
$R_{\mathrm{G,int}}$	0.4 Ω
$R_{\theta\mathrm{JC}}$ top / $R_{\theta\mathrm{JB}}$	0.2 / 1.5 K/W

- Low threshold and −4 V limit: gate drive needs care (bump, negative bias)
- Top of the package at source potential: thermal interface must insulate
- Seven interleaved stripes, pad 2 next to the gate is the natural Kelvin point

3 Submodule voltage

How many submodules? Fewer is better, until the 100 V rating

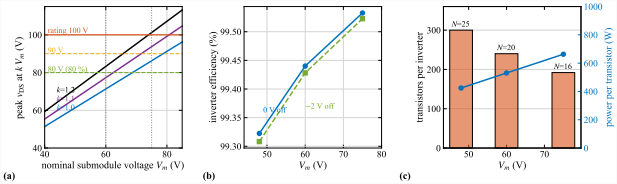


- Same phase current in every leg ⇒ conduction loss, transistors, drivers and isolated supplies all scale with N
- 50 V (N=24): 1.5 × the conduction loss and 50 % more parts than 75 V
- Upper bound: $\hat{v}_{DS} = kV_m + \Delta v_{OS}$; 75 V: 85 V nominal, 100 V at the maximum bus ($k = 1.2$)

Decision

$V_m = 75$ V, $N = 16$; no margin left at $k = 1.2$, 60 V ($N = 20$) is the fallback

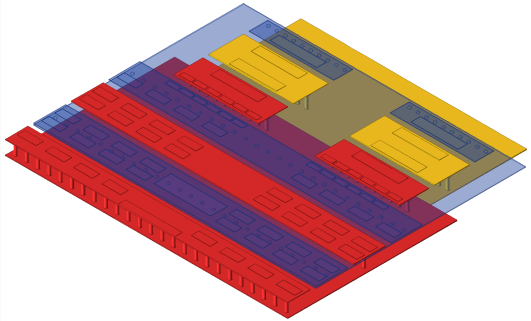
Voltage utilisation: 75 V or 60 V submodules



- 75 V nominal with $k = 1.2$ overshoot/ripple factor $\Rightarrow \approx 100$ V peak at the device
- Keeping ≤ 80 V needs $V_m \leq 57$ V ($k = 1.2$) or ≤ 69 V ($k = 1$)
- Trade-off: 60 V submodules $\rightarrow$ $N = 20$, more cells and drivers, more margin
- Rev A board designed and tested for 75 V

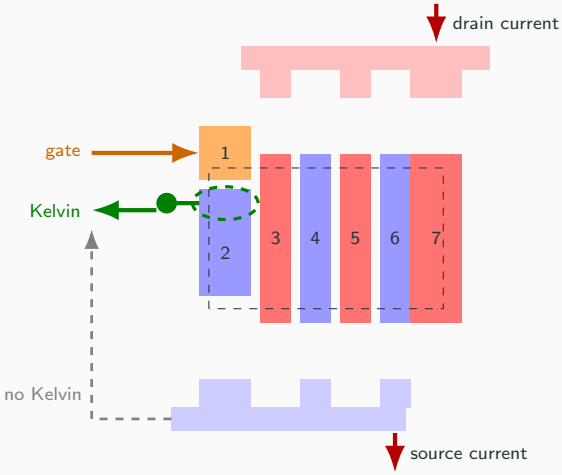
4 One device, one driver

Reference commutation cell



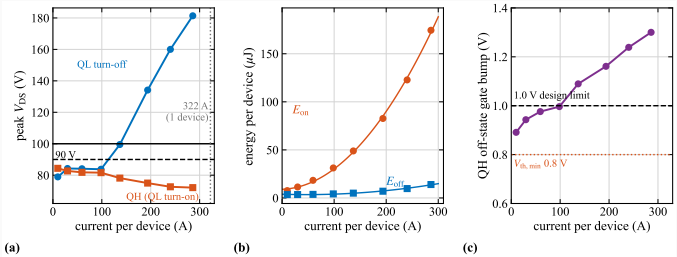
- One high-side and one low-side EPC2361 with local 0805 MLCCs at the drains
- Vertical power loop: L1 forward, L2 return 0.1 mm below
- Q3D: $L_{\mathrm{loop}} = 0.374\,\mathrm{nH}$ per cell, gate loop 1.71 nH, L_{CS} 0.030 nH with Kelvin source
- Driver per device: $2\,\Omega$ on / $0\,\Omega$ off, 5 V
- Same cell is reused for every n: only the number of identical cells changes

Kelvin source on the EPC2361



- Pad 2 is a separate source pad next to the gate pad; its gate end carries almost no load current
- Kelvin via at that end; return routed to the driver on its own path (L2 / L4)
- Gate loop (horizontal) $\perp$ power loop (vertical): minimum coupling
- Q3D: $L_{CS} = \mathbf{0.030\,nH}$ with Kelvin, 0.061 nH without $\Rightarrow \approx 0.5$ vs 1.0 V subtracted from V_{GS} at 17 A/ns

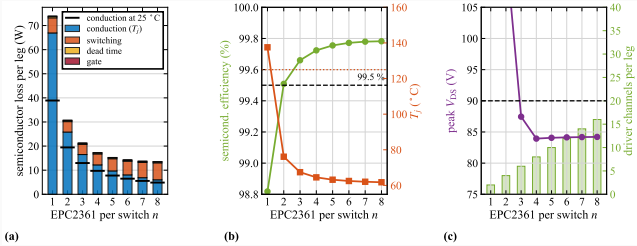
One EPC2361 per switch: how fast and how much overshoot



- LTspice double pulse at 75 V with the Q3D cell (one cell of the building block populated)
- Turn-off dv/dt grows with current: peak V_{DS} 100 V at 137 A, 134 V at 194 A per device
- E_{on} dominates and grows quadratically; the off-state gate bump passes 1 V above ≈ 100 A
- One device would carry 322 A peak: 75 V is **not feasible with one device** $\Rightarrow$ current per device must stay $\lesssim 100$ A

5 Devices in parallel

n identical cells per switch (one driver each)



- 228 A_{rms} per leg, 50 kHz, $R_{DS(on)}(T_j)$ iterated with 2.1 K/W junction–coolant per device
- $n = 1$: T_j 138 °C and $R_{DS(on)}$ +72 %; $n = 2$: 31 W, but 114 V overshoot
- $n = 3$ / 4 / 8: 21 / 17 / 14 W per leg; above 4 the gain is small while E_{oss} and drivers grow

Decision

At least **4 EPC2361 per switch** (3 is marginal in V_{DS}; 4 also splits into two equal pairs)

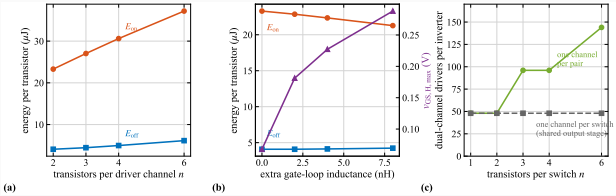
Losses vs number in parallel

n per switch	1	2	3	4	5	6	7	8
Loss per leg (W)	73.9	30.7	21.2	17.3	15.3	14.3	13.8	13.6
conduction at T_j (W)	66.9	25.8	16.5	12.2	9.7	8.0	6.9	6.0
conduction at 25 °C (W)	38.9	19.4	13.0	9.7	7.8	6.5	5.6	4.9
switching (W)	6.3	4.3	4.2	4.6	5.1	5.7	6.3	7.0
Efficiency (%)	98.82	99.51	99.66	99.72	99.76	99.77	99.78	99.78
Loss per device (W)	36.9	7.7	3.5	2.1	1.5	1.2	1.0	0.8
T_j (°C)	138	76	67	65	63	62	62	62
Peak V_{DS} (V)	181	114	87	84	84	84	84	84
Driver channels per leg	2	4	6	8	10	12	14	16

Semiconductor losses only (conduction at T_j, switching incl. E_{oss}, dead time, gate); board copper and MLCC losses are added for the block on the verification slides. Efficiency relative to 6.25 kW per leg.

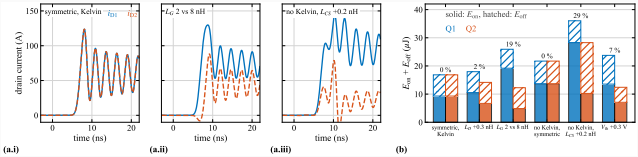
6 Gate drive: from 4 devices to a building block

Can one driver drive all four?



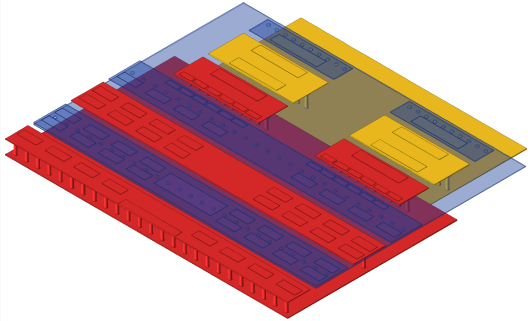
- One channel for 3 / 4 / 6 transistors: E_{on} per transistor +16 / +31 / +60 %
- Peak sink current 12–14 A, above the 8 A rating
- Longer gate tree brings the gate bump back
- One driver per device: symmetric, but 4 channels and isolated supplies per switch

Why symmetry: current-sharing sensitivity



- Static sharing is fine ($R_{\mathrm{DS(on)}}$ has positive TC)
- Dynamic sharing is set by the gate circuits:
  - gate-loop mismatch 2 vs 8 nH $\rightarrow$ 19 % imbalance, 4:1 E_{on}
  - common-source mismatch without Kelvin $\rightarrow$ 29 %, $2.8 \times E_{\mathrm{on}}$ in one device
- $\Rightarrow$ star-connected gate drive with Kelvin returns, one driver channel per pair

Building block: two mirrored cells, one centre driver



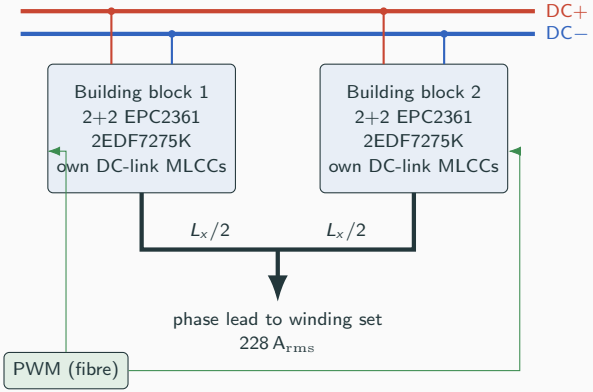
- Two cells mirrored about the driver: identical gate loops and Kelvin returns for both transistors of a switch
- One 2EDF7275K channel drives a **pair** symmetrically (star gate network)
- Each cell keeps its own commutation loop (0.374 nH per cell)

Decision

Building block = 2 × EPC2361 per switch, one driver; **4 per switch = two blocks**

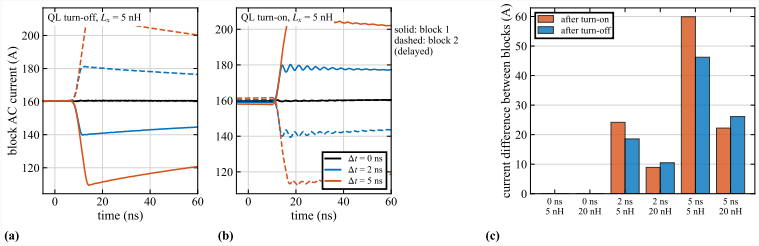
7 Two blocks in parallel

Two building blocks per leg



- Two unchanged blocks share the submodule DC link and are joined at the AC side
- Per block 114 A_{rms} (228 A per leg)
- Commutation stays inside each block (own MLCCs)
- System: 96 blocks, 96 drivers, 384 EPC2361; series DC chain carries 250 A

Sharing between blocks: driver skew and the AC connection



- LTspice, two complete blocks at 322 A, PWM of block 2 delayed by Δt; L_x = AC inductance between the switch nodes
- At each edge the leading block takes $\Delta I \approx V_m \Delta t / L_x$ and keeps it until the next edge, where the sign reverses: a circulating current at f_{sw} through the AC connection
- Δt = 2 ns: ΔI = 24 A (L_x = 5 nH) / 9 A (20 nH); Δt = 5 ns: 60 / 22 A; E_{on} 50 vs 38 µJ per block at 5 ns, 5 nH (48 vs 48 µJ at 2 ns, 20 nH)

Two blocks: requirements

Static sharing

- $R_{\mathrm{DS(on)}}$ has a positive temperature coefficient
- Identical AC and DC connections (length, cross-section) for both blocks

Dynamic sharing

- One PWM receiver, fan-out to both drivers with matched traces
- Driver skew ≤ 2 ns (2EDF7275K part-to-part delay from its datasheet still to be checked)
- $L_x \gtrsim 20$ nH between the switch nodes (phase-lead geometry), not a stiff short bar

Verification

- Double-pulse with both blocks: Rogowski on each block's AC lead (outside the commutation loop, so it fits)
- Measure ΔI for a deliberately added delay, compare with LTspice
- Continuous run: thermal camera on both blocks

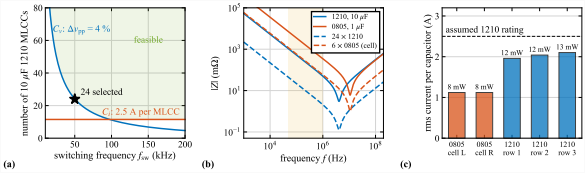
8 Verification of the building block

Building block at the 300 kW operating point

Current per block	114 A_{rms}
Conduction (T_j)	6.1 W
Switching + E_{oss}	2.4 W
Dead time + gate	0.3 W
Board copper (Q3D, incl. harmonics)	5.7 W
MLCCs	0.5 W
Total per block	**14.9 W**
Efficiency (3.125 kW per block)	**99.52 %**
Per transistor / T_j	2.2 W / 65 °C

- Pair energies from the block LTspice model (one driver per pair), copper and MLCC losses scaled with I^2 from 100 A
- $\cos\varphi = 1.0$ / 0.85: 12.4 / 16.5 W per block
- The following slides show the block analysis at its original 100 A_{rms} design point (150 kW with $M = 1.15$, $\cos\varphi = 1$)

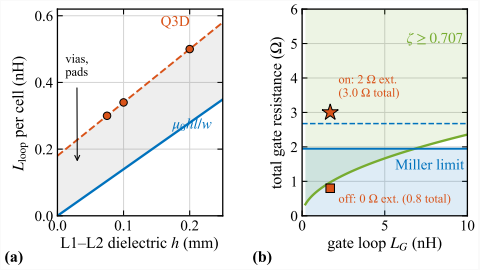
DC-link capacitors



	local	bank
Part	TDK 0805 1 µF 100 V X7S	Murata 1210 10 µF 100 V X7S
Number per block	2×6	24
C_{eff} at 75 V	0.45 µF	3.0 µF
ESR / ESL	6 mΩ / 0.48 nH	3 mΩ / 0.5 nH
Ripple per part	1.1 A_{rms}	2.0–2.1 A_{rms}
Loss per part	7.5 mW	11.5–13.3 mW

- Bank sized by voltage ripple below ≈ 50 kHz, by ripple current above
- Total capacitor loss ≈ 0.39 W per block at 100 A_{rms}

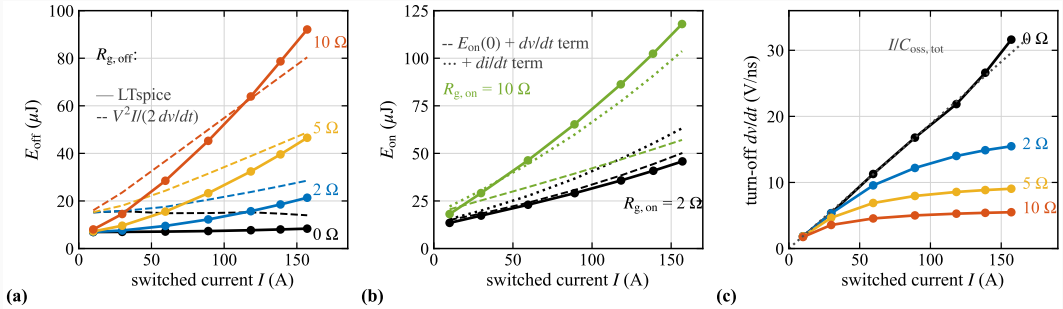
Extracted parasitics (Q3D)



Power loop per cell (copper + ESL)	0.374 nH
Gate loop L_G per transistor	1.71 nH
L_{CS}, Kelvin / no Kelvin	0.030 / 0.061 nH
Gate loop – gate loop mutual	0.46 nH
Busbar, leg to leg (1 MHz)	3.35 nH
Busbar, series tab to leg	41 nH

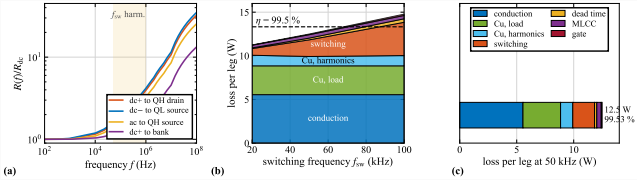
Busbar: DC+ and DC− tabs leave at opposite ends (41 nH); bring them out together or use a laminated cell interconnect.

Switching behaviour



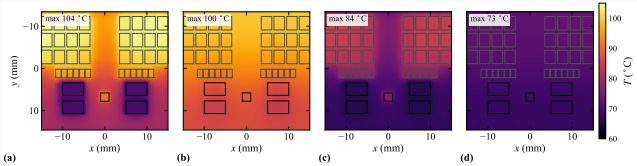
LTspice with the Q3D power loop and gate star, 75 V, $R_{g,on}/R_{g,off} = 2\ /\ 0\,\Omega$ external.

Losses and efficiency



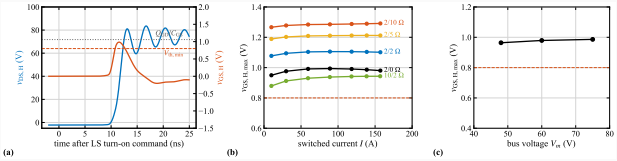
- Rated point (75 V, 100 A_{rms}, 50 kHz): **12.5 W per block**, $\eta = 99.53\,\%$
- At 300 kW ($\cos\varphi = 0.9$) the block carries 114 A_{rms}: see the previous slide
- Worst case 14.3 W (99.46 %)
- 1.91 W per transistor
- Copper and switching harmonics add $\approx 1/3$ to the DC copper loss

Thermal design



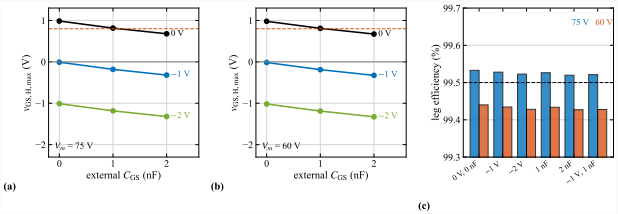
- Top-side cooling through an insulating interface to a liquid cold plate (integration concept)
- Icepak: $T_j \approx 64\,^{\circ}\mathrm{C}$ at the rated point
- Lab: fan heatsink on the Rev A board (later slide)

Gate bump at the high side



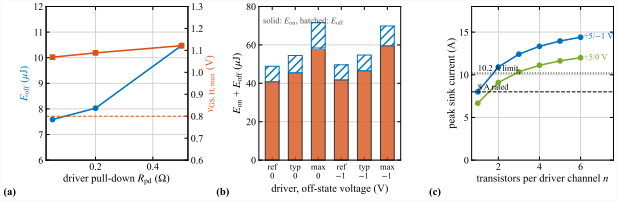
- When the low side turns on, the high-side drain sees dv/dt: Miller charge flows into C_{GS}
- Bump $\approx Q_{GD}/C_{GS} \approx 1.06\,\text{V}$, $\approx 2\,\text{ns}$, **independent of current and voltage**
- Below the 1.1 V typical threshold, close to the 0.8 V minimum: no shoot-through in simulation, but little margin

Mitigation: negative off-voltage



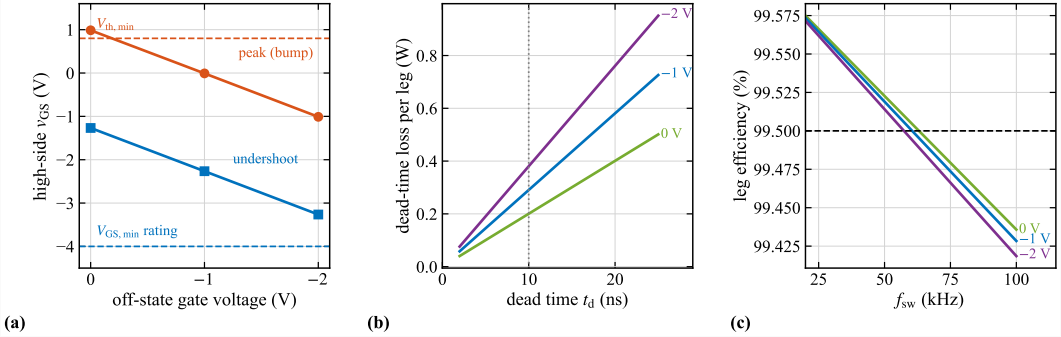
- With $-1\,\mathrm{V}$ off-voltage the peak falls from $0.99\,\mathrm{V}$ to $-0.01\,\mathrm{V}$
- Undershoot $-2.3\,\mathrm{V}$, rating $-4\,\mathrm{V}$
- Cost: reverse conduction drop in the dead time rises by $1\,\mathrm{V}$
- Dead-time loss 0.20 / 0.29 / $0.38\,\mathrm{W}$ per leg for 0 / -1 / $-2\,\mathrm{V}$
- **Chosen:** $\approx -1.24\,\mathbf{V}$

Driver: Infineon 2EDF7275K

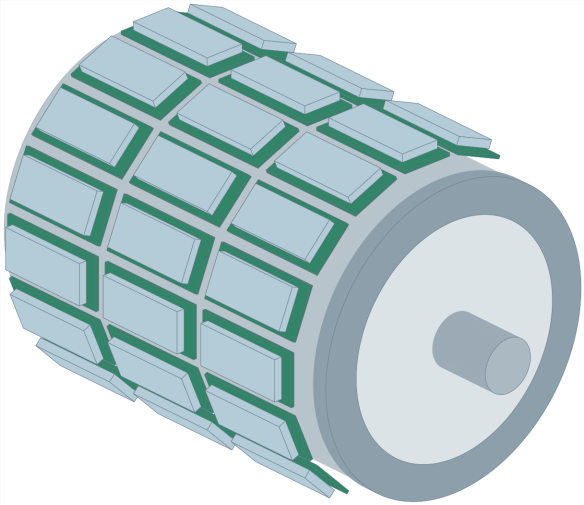


- Dual isolated channel, 4 A source / 8 A sink, 0.85 / 0.35 Ω
- 4 V UVLO: allows a split $+5/-1.24$ V supply per channel
- Sink strength sets the turn-off (E_{off}), not the bump
- LGA-13, 5×5 mm; functional isolation only

Negative rail vs dead-time loss

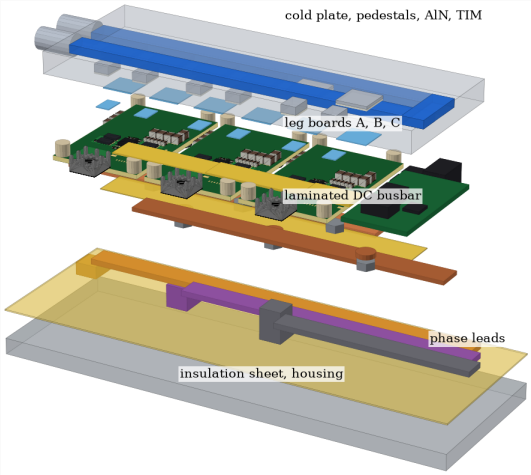


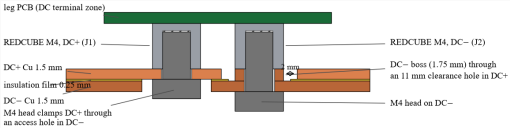
Machine-integrated drive concept



- One submodule = one cell assembly on a liquid cold plate (AlN insulation); at 300 kW six blocks (3 legs $\times$ 2)
- Laminated busbar along the cells, phase leads to the winding sets
- Machine side feeds back into N (winding sets) and f_{sw} (ripple, iron loss)

Cell assembly and busbar

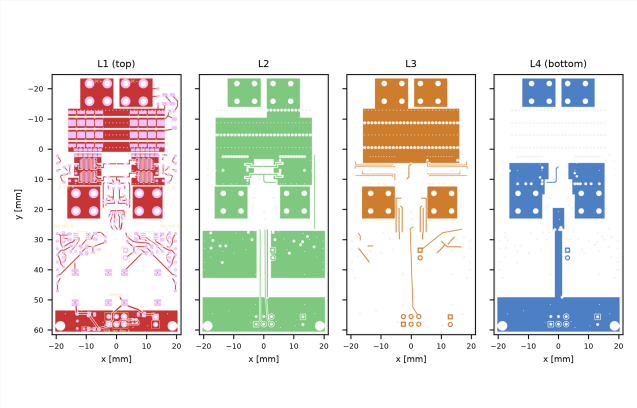




- Leg-to-leg busbar inductance 3.35 nH (Q3D)
- Series tabs: 41 nH because DC+ and DC− leave at opposite ends

9 Rev A fabrication board

Rev A board: overview

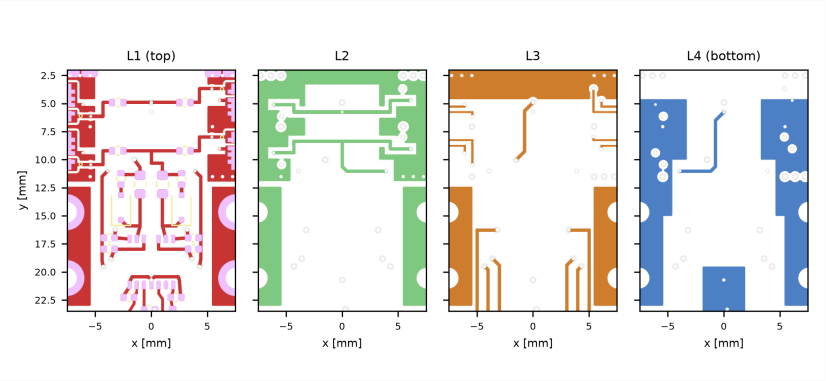


- 42×85.4 mm, 4 layers, 4×70 µm
- DC terminals (top), two cells, driver corridor, AC terminals below each switch node, gate supplies, primary side (bottom)
- Outer strips: heatsink holes, probe landings, LEDs
- Fully scripted (KiCad 7 Python), DRC 0 / 0, wired schematic identical netlist

Stack-up and design rules

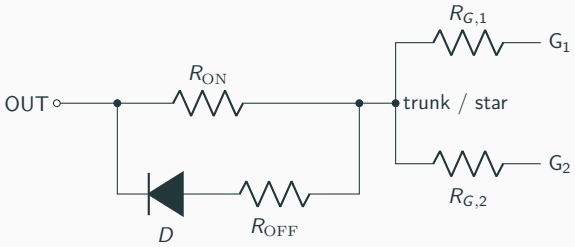
L1 70 µm	power, pads, gate stars	all components
prepreg 0.1 mm		vertical commutation loop
L2 70 µm	DC− plane	loop return, Kelvin collectors in narrow keepouts
core ≈ 1.1 mm		
L3 70 µm	DC+ plane	HS gate trunk, supply columns, probe sense pairs
prepreg 0.1 mm		HS gate trunk over its Kelvin return
L4 70 µm	AC plane	switch node to J3/J4, HS Kelvin return, primary ground
Rules	0.15/0.15 mm	power gaps 0.2 mm under solder mask (IPC-2221B B4: 0.13 mm for 51–100 V)
Vias	0.6/0.3 mm, 0.45/0.25 mm	no via-in-pad, standard process

Gate loops in the driver corridor



- HS: trunk on L3, Kelvin return directly below on L4; LS: star on L1, Kelvin collector on L2
- Kelvin collectors join on the symmetry line ⇒ identical gate loops for both transistors

Separate turn-on and turn-off paths

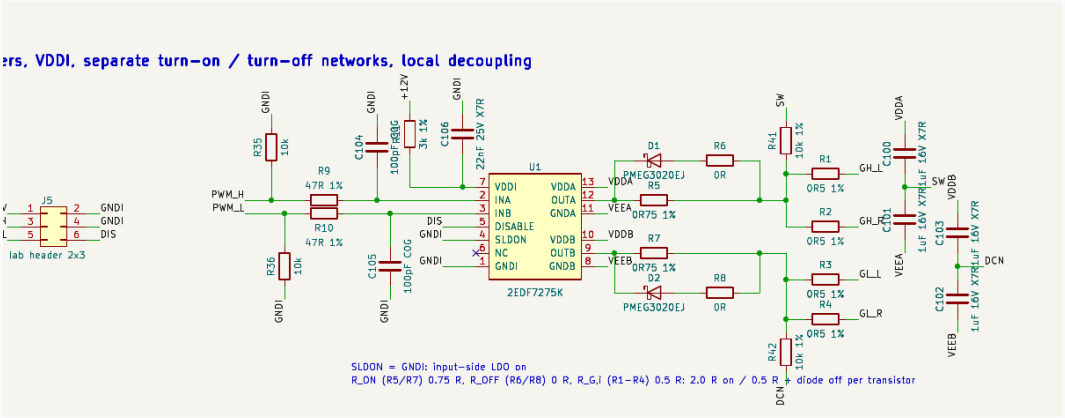


$$R_{\mathrm{on,eff}} = 2R_{\mathrm{ON}} + R_{G,i}$$
$$R_{\mathrm{off,eff}} \approx 2(R_{\mathrm{ON}} \parallel R_{\mathrm{OFF}}) + R_{G,i}\ (+V_F)$$

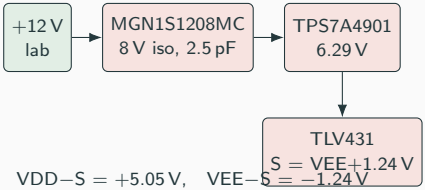
Part	default	per transistor
R_{ON} (0603)	0.75 Ω	on: 2.0 Ω
R_{OFF} (0402)	0 Ω	off: 0.5 Ω + D
D PMEG3020EJ	SOD323F	
$R_{G,i}$ (0402)	0.5 Ω	

- Peak gate current 3.1 A (on) / 6.5 A (off): within 4 / 8 A
- Same as the simulated 2 Ω / 0 Ω external
- Values changeable by population
- 10 kΩ pull-down star → Kelvin per switch

Driver schematic (KiCad)

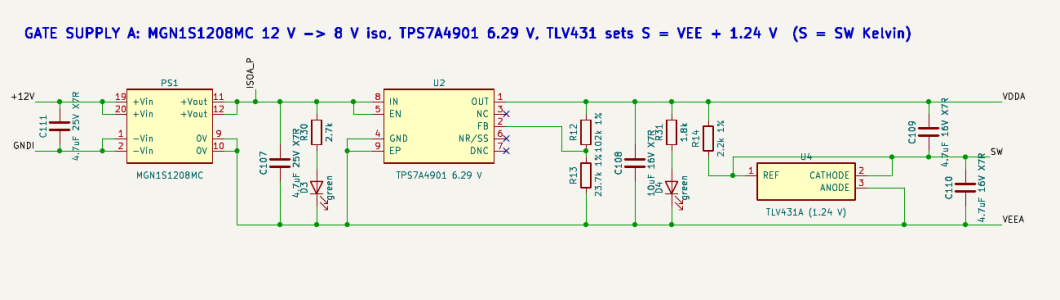


Isolated gate supplies



- Two isolated supplies: HS reference = switch node, LS reference = DC− (floats up to 1.2 kV in the stack)
- TPS7A4901 replaces LP2951: LP2951 has no VSSOP and its legacy chip needs ESR $\geq 30\,\mathrm{m\Omega}$
- $V_{\mathrm{OUT}} = 1.185\,\mathrm{V}\,(1 + 102\,\mathrm{k}/23.7\,\mathrm{k})$; stable with ceramics $\geq 2.2\,\mu\mathrm{F}$
- TLV431 stable: $> 5\,\mu\mathrm{F}$ on S−VEE (unstable band 0.02–0.8 µF)
- Budget per channel at 50 kHz: 65 nC, 7.6 mA, 61 mW

Supply schematic (KiCad)



Isolation domains

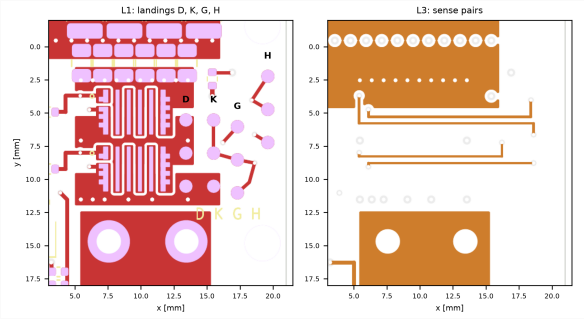
Domains	outer	inner
DC+ – switch node	0.20	0.65
DC+ – DC−	0.70	1.07
HS – LS	0.20	0.40
HS – primary	1.37	1.08
LS – primary	1.50	0.75

minimum copper

spacing [mm] on the routed board

- Power domains $\leq 100\,\mathrm{V}$: 0.2 mm under mask
- Primary side separated functionally (lab board)
- In the 1200 V stack: PWM via fibre / reinforced isolator and isolated 12 V at the controller side; 2EDF7275K is functional only

Double-pulse test: measurement landings



- Probe pads only, outside the commutation loop; spring ground at 2.5 and 5.0 mm
- **D**: V_{DS} of Q3 / Q4 at the package
- **K**: v_{KS} = Kelvin vs power-source copper
- **G**: V_{GS} of Q3 / Q4 (sense pairs on L3)
- **H**: V_{GS} of Q1 / Q2 (isolated probe)
- Inductor current: Rogowski / Pearson off the board

E_{on}, E_{off} and per-transistor current

- $E = \int v_{DS}\, i_D\, dt$; E_{off} with $i_D = I_L$; E_{on} with I_L plus the high-side output charge $Q_{\mathrm{OSS}}\, V_{\mathrm{DC}}$
- Deskew $< 50\,\mathrm{ps}$: 100 ps at 50 V/ns and 100 A $\approx 0.75\,\mu\mathrm{J}$ error ($\approx 8\,\%$ of E_{on})
- Per-transistor current without a shunt (no extra loop inductance):

$$v_{KS} = L_s \frac{di_D}{dt} + R_s i_D \quad \Rightarrow \quad \Delta i_D(t) \approx \frac{1}{L_s} \int v_{KS}\, dt$$

  $L_s \approx 0.03\,\mathrm{nH}$: $\approx 0.5\,\mathrm{V}$ at 17 A/ns; scale with $i_3 + i_4 = I_L$, ratio gives the dynamic sharing
- Same principle as stray-inductance di/dt sensing in SiC short-circuit protection (Sun, CPES 2019; Xue et al., TPEL 2020)
- Rogowski holes in the cells: not possible without opening the L1/L2 loop; coil bandwidth (20–50 MHz) too low for 2–3 ns edges

Bring-up: test points, LEDs, protection

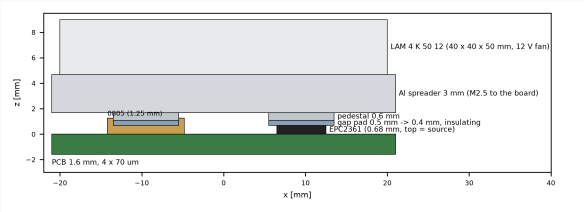
Item	nets	expected
TP15, D7	+12 V	12 V present
TP9/12, D3/5	ISO / VEE	MGN1 output 8–10 V
TP10/13, D4/6	VDD / VEE	6.29 V
TP11/14	S / VEE	1.24 V
TP16, D8 (red)	DC link	charged; 100 k bleeder
R35/R36	PWM	10 k pull-downs
R41/R42	star–Kelvin	10 k gate pull-downs
J7	+12 V	fan

Power-up order

1. 12 V (current limit), LEDs on, TP values
2. PWM without DC link: driver outputs on R5/R7, gates on R1–R4
3. DC link 12 → 24 → 48 → 75 V, double-pulse test
4. Heatsink, continuous operation

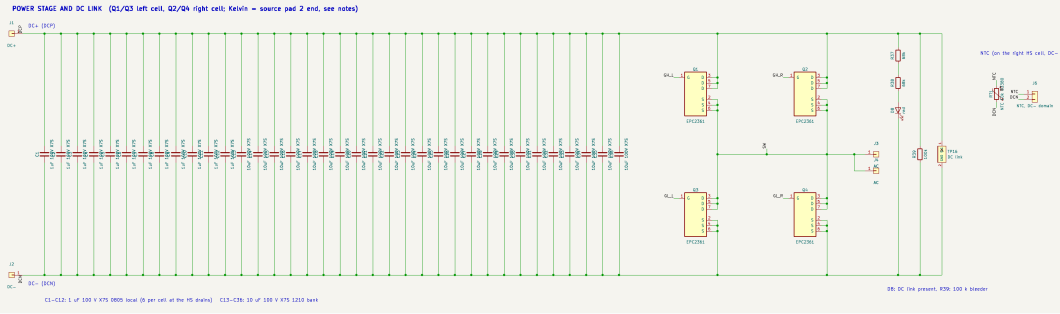
Power down in reverse: DC link first, then 12 V.

Cooling for the load test



- Fischer LAM 4 K 50 12 (0.65 K/W, 12 V fan from J7), off the shelf
- Al spreader $42 \times 14 \times 3$ mm (workshop part), $4 \times$ M2.5
- Insulating gap pad 0.5 mm, ≥ 5 W/mK: 5.3 K/W per transistor (dominant)
- $T_j \approx 57\,^{\circ}$C (40 °C ambient); flat plate + 1 mm pad: $\approx 70\,^{\circ}$C
- Next step: AlN insulator (≈ 1 K/W per transistor)

Power-stage schematic (KiCad)



Status and next steps

Verification status

Done	DRC 0 violations / 0 unconnected; wired schematic = PCB netlist (37 nets); 3D models for all parts, STEP export; Gerber, drill, pick-and-place, BOM (35 lines, 109 parts + mechanics); datasheet checks: TPS7A4901 pins/stability, TLV431 pinout and stability, PMEG3020EJ; spreader drawing; gap pad proposed (Bergquist TGP 5000, 0.5 mm)
Prepared	Q3D re-extraction of the routed copper (`run_bb_q3d_fab.cmd`); needs the KTH licence (campus / VPN)
Open	Q3D result vs 0.374 nH design value; gap-pad breakdown rating; fab-house review of 0.2 mm gaps; isolation concept for the 1200 V stack; two-block parallel connection (AC/DC interconnect, PWM fan-out)

Next steps

1. Re-extract the routed board in Q3D, update loop and gate-loop values
2. Order Rev A, assemble, bring-up (12 V, PWM, then DC link)
3. Double-pulse test at 12–75 V: E_{on}, E_{off}, gate bump, sharing (D, G, K, H landings)
4. Continuous operation with the fan heatsink; thermal camera
5. Two blocks in parallel (300 kW leg): double-pulse and continuous test, block-to-block sharing with a Rogowski on each AC lead
6. Compare with LTspice / Q3D; decide 75 V vs 60 V submodules
7. Rev B: cell interconnect with laminated tabs, AlN interface, integration form factor (36 mm pitch)

Thank you

Questions?